

Homework 9: Searching Ordered Data and Search Trees

Data Structures

Write pseudo-code not Java for problems requiring code. You are responsible for the appropriate level of detail.

For questions 1 – 2, compare the efficiency of using sequential search on an ordered table of size n and an unordered table of the same size for the key *target*:

- If no record with the key *target* is present.
 - If one record with the key *target* is present and only one is sought.
- If more than one record with the key *target* is present and it is desired to find only the first one.
 - If more than one record with the key *target* is present and it is desired to find them all.
- Write a method `delete(key1, key2)` to delete all records with keys between `key1` and `key2` (inclusive) from a binary search tree whose nodes look like this:

Left	key _{<i>i</i>}	right
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- Write a method to delete a record from a B-tree of order n .

p ₀	r ₁	p ₁	r ₂	p ₂	r ₃		p _{<i>n</i>-1}	r _{<i>n</i>}	p _{<i>n</i>}
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